Maastricht University

Global and Local Energy Consumption

Changes in energy consumption Between 1985 and 2024

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MAT2007 – Introduction to Programming

(Disclaimer: Not all results could be included for this report, for the complete results see the github link [1])

October 2026

1 Introduction

The use of energy has more than doubled in the last 39 years. Information about how this energy consumption shifted is essential to improving durability and efficiency. The goal of this project is to investigate the global and local consumption of energy between 1985 and 2024. Furthermore, the distribution of this energy will be discussed and the difference in usage for specific regions and countries between 1985 and 2024. The research question is: *How have global and local energy consumption and energy mixes changed from 1985 to 2024?*

2 Dataset

For the analysis the energy dataset from Our World In Data (OWID) was used, which contains over 100 different trackers for variables such as energy consumption, type of energy and energy production. These variables were tracked for virtually every nation on earth as well as for specific regions and groups. The relevant columns for our analysis were: country, year, and energy consumption per type.

3 Analysis

The analysis of the data was done in python using numpy and pandas. Firstly, the dataset was filtered to only contain countries and years between 1985 and 2024, the reason for this being individual countries could be manipulated more easily, and this specific period was chosen because of the lack of data outside this range especially for smaller countries. The analysis itself consisted of the following steps:

- Calculated each country's share of global consumption
- Plotted global energy mix over time with groups: Fossil, Renewable and Nuclear
- The percentage points changes for each group were calculated for 76 countries between 1985 and 2024
- Calculated mean percentage point changes for specific regions

4 Results & Discussion

Global energy consumption increased by 109.21% between 1985 and 2024, although this growth was distributed unevenly across countries. China was the largest increaser among the countries examined, with an increase of 690.30%. In comparison, consumption in the United States increased by only 31.02%. This difference is also seen in their contributions to global energy consumption: China's share went from 7.59% in 1985 to 28.68% in 2024. These results show a significant shift in the distribution of global energy consumption, with China surpassing the USA by far in 2024.

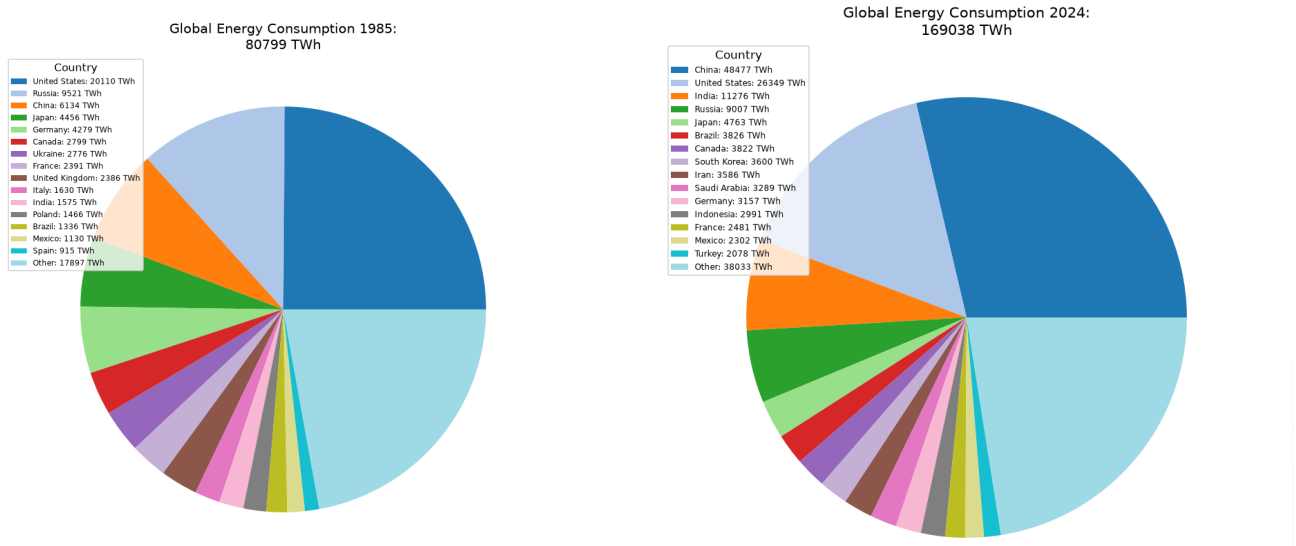


Figure 1: Global energy distribution 1985 & 2024.

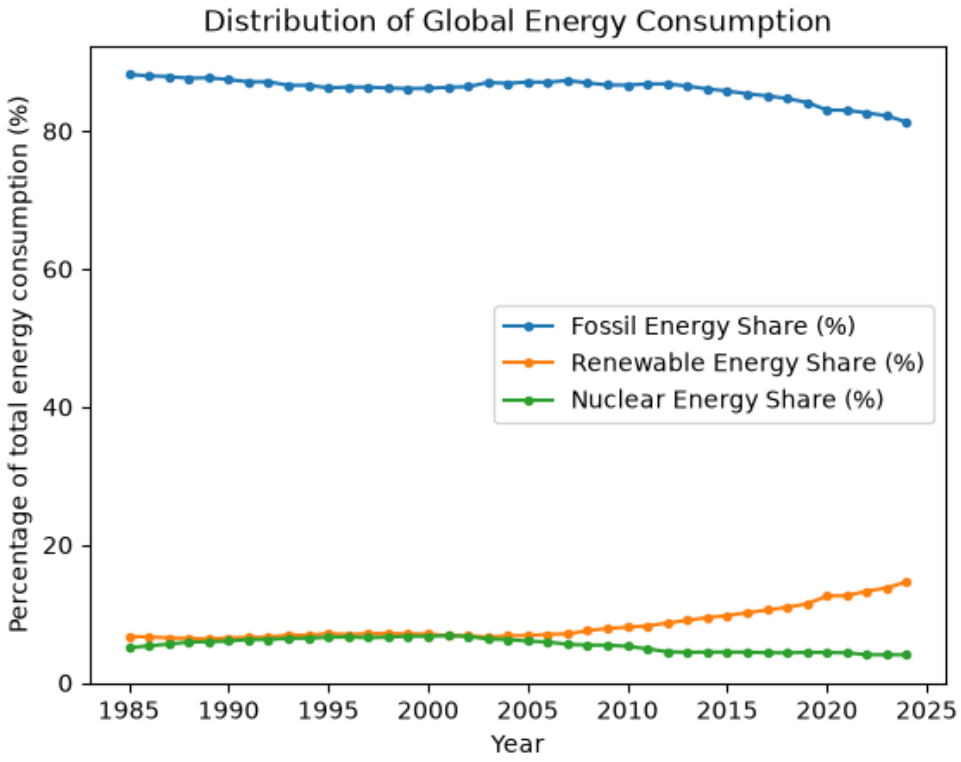


Figure 2: Global energy mix between 1985-2024.

As can be seen from the figure fossil energy consumption has decreased (by -6.91 pp) and renewables have increased (by +7.94 pp), while nuclear only dropped by -1.03 pp. These changes, according to climate organisations are not significant enought to slow down climate change.

Looking at indivdual countries there is a general trend in decreasing fossil fuels consumption with Denmark being the leader with a -41.57 pp decrease. There are a few exception like Taiwan, and Canada who have increased fossil consumption, most of these have specific reason though; Taiwan was compensating their loss in nuclear energy by increasing fossil etc. Nuclear energy has decreased overal but is polarized: some countries like Czechia have increased their consumption from this source (17.51 pp) while others like Taiwan have decreased it.

Table 1: Mean change in energy shares, 1985–2024 (pp)

Energy source	Global	Europe	East Asia	North America
Fossil fuels	-8.83 ± 9.94	-17.53 ± 7.97	-2.10 ± 11.69	-2.65 ± 6.65
Renewables	$+7.95 \pm 8.88$	$+15.34 \pm 7.99$	$+5.27 \pm 4.95$	$+1.72 \pm 5.40$
Nuclear	$+0.88 \pm 5.24$	$+2.18 \pm 7.15$	-3.17 ± 9.12	$+0.93 \pm 1.34$

(Uncertainties were calculated as the standard deviation between countries.) As can be seen from the uncertainties table 1, even we have on average switched more to renewables, the variation between countries is enormous highlighting the fact that there is still a lack of common policy and cooperation between regions and within them.

5 References

[1] Coolen, C. (n.d.). MAT2007 project. GitHub. https://github.com/casco06/MAT2007-Project

[2] Our World in Data. (n.d.). Energy data [Data set]. GitHub. https://github.com/owid/energy-data